

CD Moisture Profile and Reel Scanner Comprehensive Summary

N. Costello 08/02/2024

Background

APD scanner out of service waiting on parts as of July 8th. In interim the PM team was relying solely on the Reel Scanner to control their sheet moisture.

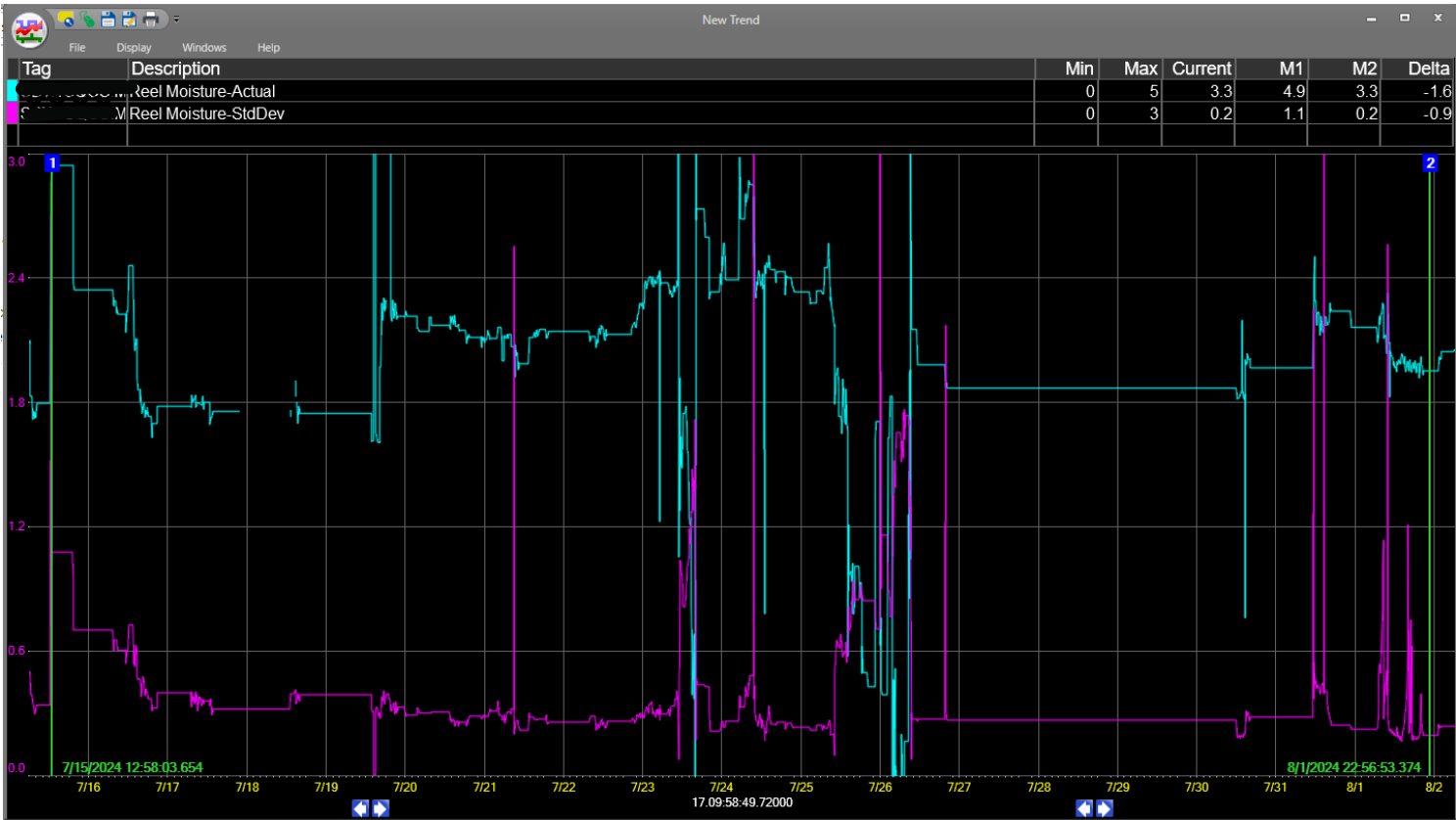
On July 12th, the team sped up the machine following the East Bale Handling Conveyor upgrade but failed to compensate TAD temps for moisture resulting in loose wound and shifting rolls to Converting. Moisture was then corrected morning of July 13th and appeared stable at target speeds.

The team completed a stock off CIL on July 15th. Stock off event showed poor sheet control with the sheet following back to the former when trying to start up, indicating another potential moisture issue. It was identified the steam box temperature and pressure control loop was failing to control properly resulting in the steam box kicking out without warning. This resulted in moisture variance from roll-to-roll. The steam box condensate return valve was put in “manual mode” later this day. The rejected rolls from Converting for July 15th and 16th aligned with events where the steam box unknowingly kicked out.

On July 16th the reel scanner was put into “manual mode” after lab samples indicated the reel scanner was reading 3.0-3.5% less than the actual roll moisture. As of July 16th, [REDACTED] had no functioning moisture or basis weight scanners. Troubleshooting and repair began later this day. Additional rolls were rejected from Converting for loose wind and shifting.

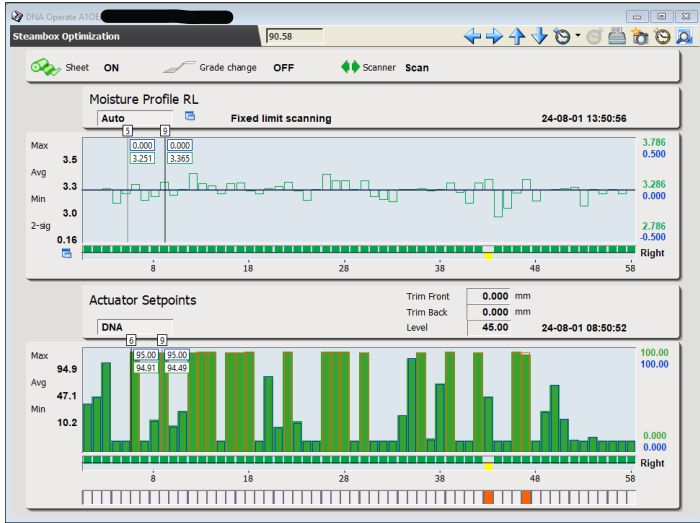
Reel Scanner & CD Profile Executive Summary

- Below is a trend 07/16/24-08/02/24 showing our reel scanner moisture %, and 2-sig values. Marker (M1) indicates pre-CD profile work, and marker (M2) shows post-CD profiling.

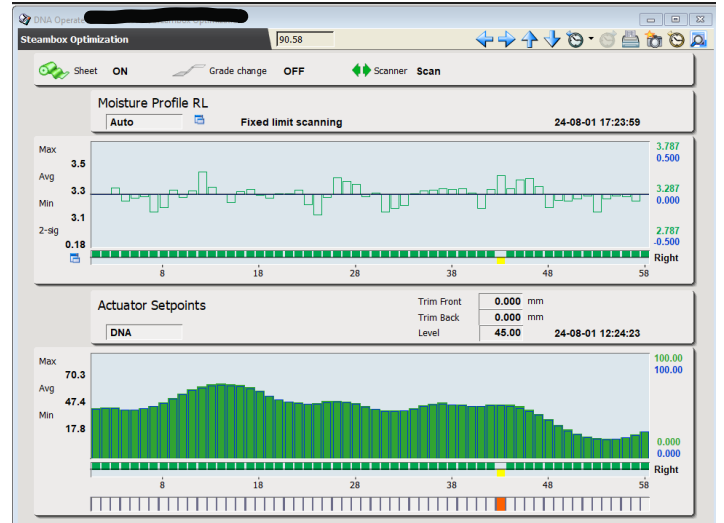


Reel Moisture Output and Variance Pre/Post Mapping Indicated by Trend Markers.

- Below is the steam box actuators pre/post-dilution profile mapping. The steam box is now being controlled by the reel scanner as the APD scanner remains out of service pending parts arrival. This control method had to be programmed, as the reel scanner historically has never been used to control this loop.



Pre-DP Mapping: Multiple Steam Box Actuators Maxed Out Limiting Drying Capability in CD Direction.



Post-DP Mapping: Increased Range, Reduced Median Valve Position & Moisture 2-sigma at 0.18.

- Below are the new mapped DP actuator positions, and a trend of reel moisture and TAD2 exhaust temperatures after mapping. The reel moisture + RL 2-sigma, basis weight + BW 2-sigma, TAD2 exhaust temperature, and steam box actuators remained stable overnight.



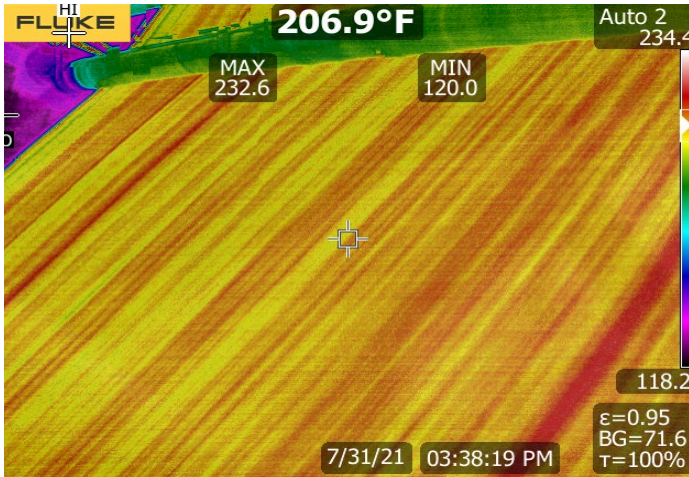
08/01/24 New Mapped Dilution Profile.



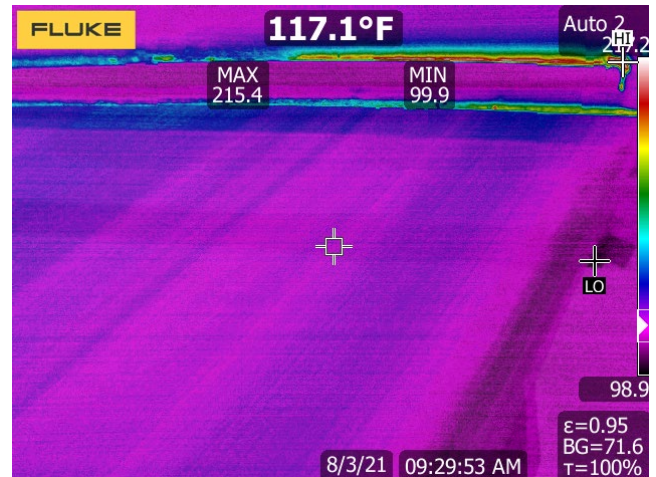
Reel Moisture and TAD2 Exhaust Temperatures Remained Stable.

- The dilution profiling map and steam box maps will be saved, and dilution profiling loop run in *manual* until reprogramming is confirmed completed and effective.
- The improvements made were confirmed in FLUKE images of the sheet at the direct outfeed of our steam box and TAD2 exit roll. There is visibly less frequent/less aggressive streaking (please note the identified

highs/lows are not accurate as the FLUKE was permitted to drift off-sheet while the images were being captured).



Pre-Mapping. Visible and Aggressive Streaking.



Post-Mapping. Less Streaking and Improved Uniformity.

Reel Scanner Synopsis

6. After re-routing blow-box air to the water jet beam to improve cog belt and beam housekeeping, the air to the reel scanner was inadvertently increased from 2,000 cfm to 2,800 cfm. This resulted in “floating” the sheet over the scanner sensor, and negatively impacted reel scanner output values.
 - a. The air lines have been disconnected from the reel scanner, as the increase in air was also prohibiting accuracy and repeatability testing, and air scanning during troubleshooting.
 - b. The team will evaluate the need for re-introducing air to the scanner, based on the below recommended settings.



Location of Air Re-Routed for Water Jet Housekeeping Improvements.

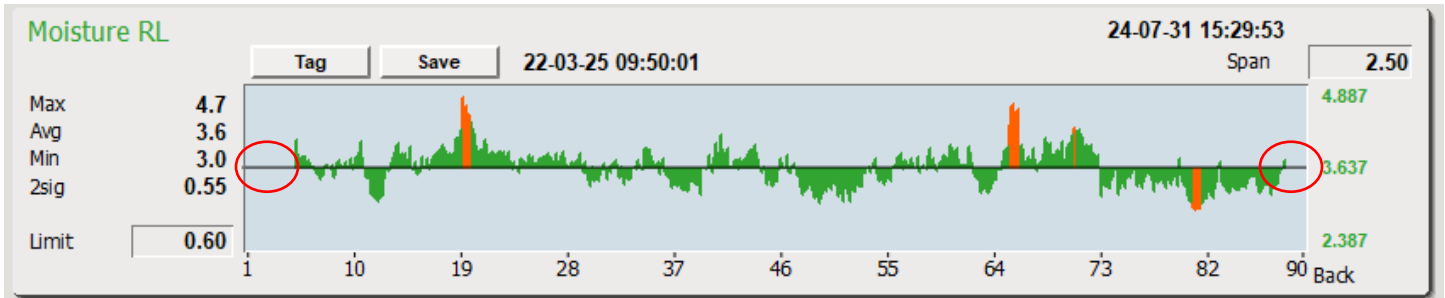
Dry, oil-free compressed air, 600 kPa
Supply coupling G 1/2"

Unit	Typical
Caliber measurement	40 lpm
Basis weight measurement	0
Purge Air	60 lpm
Air nozzles of the platform	600 lpm
Total	700 lpm

Recommended Compressed Air Settings.

7. It was also identified the Coanda nozzles were not functioning, so we did not have adequate pull to keep the sheet at the sensor head. This has been corrected.
 - a. E&I is putting a PM/SMP in place for period inspection and functionality testing.

8. We determined the reel scanner was not scanning the full sheet, so we were not getting a full CD profile. As pictured below there was data missing from both tending and drive side edges, which equates to approximately 12" of the sheet profile missing.
 - a. The front and edge trim positions were corrected in DCS. We are now scanning the full sheet except for a small region of the front edge. When setting the scanner to come all the way to the front edge, it impacts the new thread up chute. If a small degree is shaved off the front end of this chute, we should be able to capture the remaining 2-3 inches.
 - b. We are currently using "Fixed Positions" for the front and back edge; [REDACTED] recommends using the photocells that appear to be in proper working condition. This tool is more accurate as it adjusts to any variance in front or back edge positions (edge flaring, trim length etc.). However, we cannot use this method of scanning as the scanner head contacts the thread up chute as pictured.



Missing Data Due to Improperly Set Edge Detection.



Post-DP Mapping 1RL Scanner Contacts Front Edge of Thread Up Chute When Scanning Full Sheet.

Setup data		Parameter Bank
*Prof.cell pos.back	34.0	in
*Prof.cell pos.front	249.0	in
*Fixed limit back	38.5	in
*Fixed limit front	239.0	in
*Alarm limit back	5.5	in
*Alarm limit front	247.0	in
*Seek Advance	0.0	in
*Edge overscan back	7.87	in
*Edge overscan front	3.94	in
*Offsheet position	6.0	in
*Standardize pos.	6.0	in
*Fixed position	178.7	in

New RL Scanner Set-Up Data w/ Appropriate Edge Limits.

9. We determined that an incorrect sheet width was being calculated in the logic due to the fixed edge positions being set incorrectly. It appears upon install; someone forced a value of "204". This is now calculating correctly, and the scanner sees our sheet width is 205".
10. We decided to switch our moisture unit of measure calibration coefficients to "Cellulose Response" rather than "Fiber Linearization" as the cellulose uses #/3,000 sq ft. which mirrors our basis weight UOM.

Moisture Calibration Coefficients		
Water Response	Sensor	Application
Fiber_a0	0.0000	0.0000
Fiber_a1	0.2772	0.2772
Fiber_a2	-1.2772	-1.2772
Fiber_a3	1.0000	1.0000
Fiber_a4	0.0000	0.0000
Cellulose Response		
Fiber_b0	0.0000	0.0000
Fiber_b1	0.6040	0.6040
Fiber_b2	-1.6040	-1.6040
Fiber_b3	0.0000	0.0000
Fiber_b4	1.0000	1.0000
Linearization		
Fiber_d0	-0.8929	-0.8929
Fiber_d1	12.4499	12.4499
Fiber_d2	0.0000	0.0000
Linearization	FBL	
Fiber Coef A	0.0000	
Fiber Coef B	1.0000	
Fiber Coef C	0.0000	

Cellulose Response Sensor and Application Calibration Coefficients Have Proven Effective.

11. We had issues collecting sufficient data to move forward with calibration work as the scanner defaults to “Manual” and stays off-sheet following a sheet break event. A visual alarm was created and added to DCS Overview page. We were able to keep the scanner online for a 24-hour period following creation of this alarm so we could began working on our CD profile.

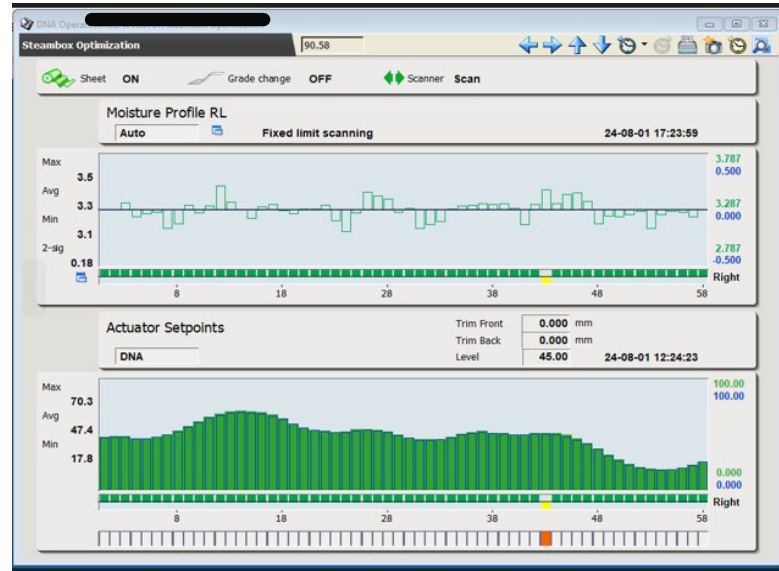


“RL SCAN NOT RUNNING” alarm created and added to overview page to help remind CRO's to return the scanner to "Automatic" and put it back on the sheet so moisture and basis weight data can be collected.

12. The scanner was calibrated with new slopes and offsets to continue running the steam box control loop in automatic. New grade profiles were configured so the slopes and offsets can be fine tuned at future grade changes. This will need further work.

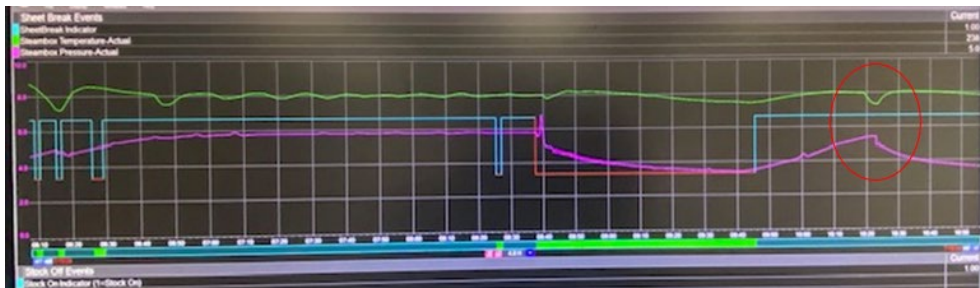


Dilution Profile Map.



Box Actuator Positions After DP Mapping. Gained 30+ points in range and reduced median value.

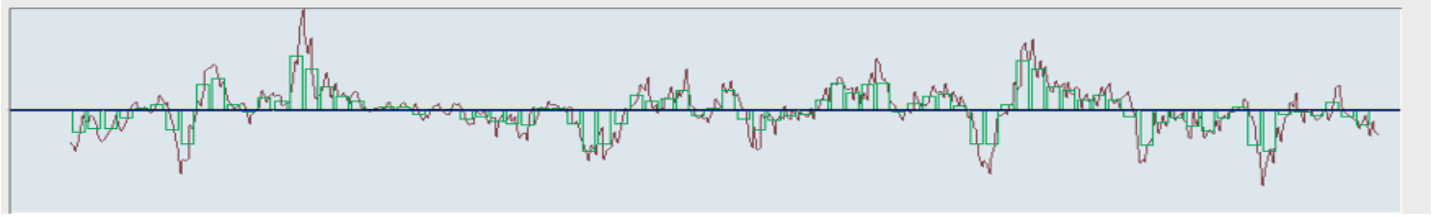
5. It was also noted that when running the steam box in Automatic, the temperature would drastically decrease while pressure simultaneously increased. The condensate return valve after the pump has a 30 second timer on it but was not quick enough to correct and the steam box would “kick out” or go to a raised position without any indication it failed. An example of this trend can be seen below.



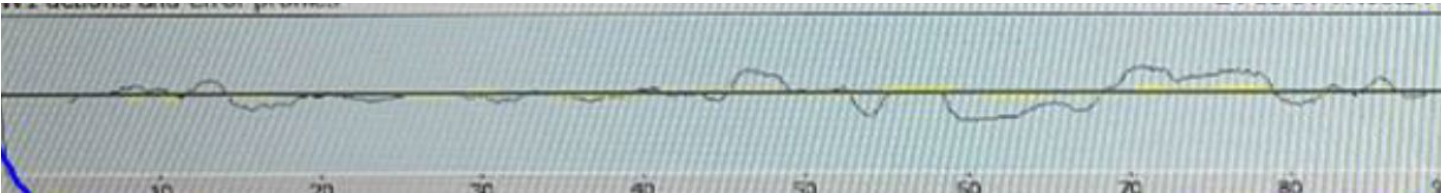
Trend of Steam Box Temperature Drop and Pressure Increase. Followed by the Steam Box Kicking Out Without Alarm.

- a. This also impacted the machine’s ability to control the CD profile. It’s been suggested that the above work will mitigate the temperature drops and inverse pressure spikes. However, the condensate return valve will run in manual until this theory proven and corrected.
- b. To help control this going forward, the Optim Factor was reduced to limit how quickly the steam box actuators can fluctuate at any given point in time. This should help keep the actuator level below its current set-point.
- c. The “Reset Shape” and “Measurement Acceptance” ratios and parameters were also adjusted to allow for less aggressive changes, which should alleviate more of our CD Profile variance.
6. Data from the scanner running over the past 12 hours shows the steam box control loop and reel scanner are stable, and the CD profile 2-sigma has been reduced from 1.1-1.5% to 0.16-0.2%.
7. We spent time troubleshooting the dilution profiling loop not controlling basis weight in Automatic. The troubleshooting process interrupted dilution profiling multiple times and caused several sheet breaks, but we were able to determine root cause of this loop not functioning.
 - a. The oven dried profile from DCS pg. 90.56 should be communicating real time data to a second page in DCS.
 - b. This second page is responsible for receiving this real-time data and communicating movement to the dilution profiling actuators.

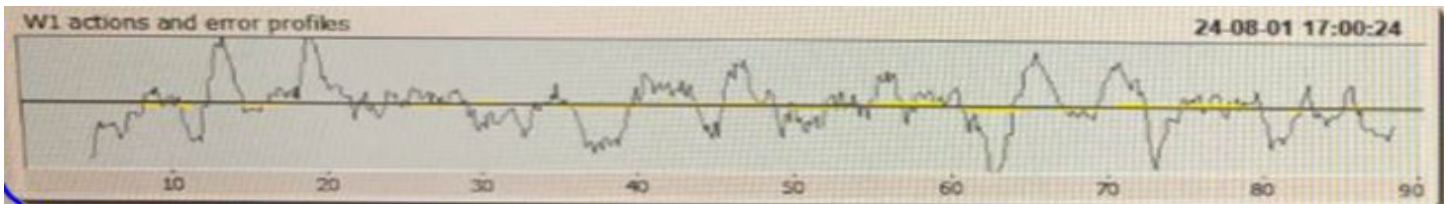
- c. The oven dried profile was keeping a “ghost file” updated, however was not keeping this second page updated properly. Therefore, the longer the reel scanner runs the more the Tuning page loses its “shape”, and the dilution profile actuators continuously drift resulting in significant basis weight swings.
- i. Below are screenshots of the Oven Dried scanner profile in real-time, and the contrasting “Tuning” profile that gives direction to the dilution profiling actuators.



Oven Dried Scanner Profile in Real-Time.



Corresponding Profile Tuning Page Showing Contrasting Profiles Due to Poor Data Transfer.



"Ghost File" Updated Which Does Accurately Reflect Our Oven Dried Scanner Profile.

8. [REDACTED] it is working to correct this programming remotely and will set a conference call up with me once complete so it can be tested.

Path Forward

- Save new steam box and dilution profile maps over oldest existing in DCS library.
- Return steam box condensate return valve to automatic and observe performance.
- Re-print all reel scanner moisture DCS pages so physical copies are available as back-ups.
- Put a new standard in place to better control how and why new maps are created.
- Finish setting up new grade profiles at changeovers.
- Implement new reel scanner housekeeping standard and complete frequent spot checks.
- Remove air lines if they are not needed. If they are needed, ensure analog gauges and appropriate flow controls are implemented so they can be centerlined for periodic inspection.
- Perform another round of steam box and dilution profiling bump testing to correlate 4-5 more data boxes to specific actuators.
- Create and implement a standard around 2-sigma basis weight and moisture CD profiles.
- Conversations with crews regarding importance of keeping the reel scanner on-sheet so data can be collected, and more valuable calculations can be made by the program.
- Create procedure for switching steam box control between APD and Reel Scanner.
- Repair sticking reset button on drive side dry-end cabinet.

- Implement PM and SMP for checking Coanda nozzle functionality.
- Continue diagnostics and re-programming of dilution profile control loop with [REDACTED] team remotely.

Additional Recommendations

- In speaking with [REDACTED] they offer IR cameras that can be tied into our sheet defect system. This will provide visual aid to the Paper Machine team and allow for easier identification of CD profile deficiencies. In addition, this is a good contingency in the event anything obscure happens to the scanners again. I am suggesting placement near the TAD/Yankee transfer point.
 - We should be able to get a good enough angle that will allow visibility to the sheet coming around the pressure roll, as well as it's completed transfer to the can.
 - This would allow us to also use this camera for easier identification of any coating or fiber bands, or blade profiling issues.
- In addition, [REDACTED] came on-site and grabbed demo footage of the sheet pulling off the blade at the Yankee. This can be added to our system via a pin-point lens. This will allow CRO's live views of the sheet performance at the creper so they can react in a timelier manner to any potential loose wind/roll shifting scenarios that may occur.



Resources

External

- [REDACTED]
- [REDACTED]
- [REDACTED]
- [REDACTED]
- [REDACTED]
- [REDACTED]

Internal

- [REDACTED] - E&I
- [REDACTED] - Sr. Process Control Engineer
- [REDACTED] - E&I Planner
- [REDACTED] - Shift E&I
- Nicole Costello - Process Engineer II
- [REDACTED] - Shift E&I

DCS Page and Logic Changes Made

- 1) Finished programming of reel scanner to steam box control loop. Created DCS pg. 90.58. Modified DCS pg. 97.55.1.21.
- 2) Re-enabled “select/deselect” steam box actuators, and option to change actuator positions for bump testing. Modified DCS pg. 97.55.1.19.
- 3) Created an alarm for “RL SCAN NOT RUNNING”, which was added to DCS pg. 90. The alarm loop number is TM1-PFC-RLwarn.F.
- 4) Reel edge positions were not visible from DCS pg. 90.70.1.2 which is used for scanner fixed point sampling and calibration. They are now visible as intended.
- 5) The sheet width was not calculating correctly. The logic blocks in loop TM1-PFC-RLsh were adjusted, and DCS pg. 97.2.1.1 was updated.
- 6) Updated DCS pg. 90 so the “running” grade and “next grade” were visible as intended.

Timeline of Events

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REDACT